Week 7 – SQL Query Implementation for E-Commerce Database

Scenario:

The Photo Frame Shopping System has collected customer, product, order, and inventory data. The management team wants to retrieve useful information from the database for daily operations and basic business analysis.

Students will apply SQL queries to search, filter, sort, and generate reports from the existing Photo Frame Shopping System Database.

Tasks:

1. Display all customer details.



```
SQL> SELECT *
  2  FROM CUSTOMER;

CUSTOMER_ID CUSTOMER_N EMAIL                    PHONE
----------- ---------- -------------------- ----------
ADDRESS
--------------------------------------------------------------------------------
REGISTRAT
---------
        201 ARUN       ARUN@GMAIL.COM       9876543210
CHENNAI
20-SEP-26

        202 PRIYA      PRIYA@GMAIL.COM      9876543211
MADURAI
21-SEP-26

CUSTOMER_ID CUSTOMER_N EMAIL                    PHONE
----------- ---------- -------------------- ----------
ADDRESS
--------------------------------------------------------------------------------
REGISTRAT
---------

        203 KARTHIK    KARTHIK@GMAIL.COM    9876543212
TRICHY
22-SEP-26

        204 DIVYA      DIVYA@GMAIL.COM      9876543213
THANJAVUR

CUSTOMER_ID CUSTOMER_N EMAIL                    PHONE
----------- ---------- -------------------- ----------
ADDRESS
--------------------------------------------------------------------------------
REGISTRAT
---------
23-SEP-26

        205 SURESH     SURESH@GMAIL.COM     9876543214
SALEM
24-SEP-26

        206 MEENA      MEENA@GMAIL.COM      9876543215

CUSTOMER_ID CUSTOMER_N EMAIL                    PHONE
----------- ---------- -------------------- ----------
ADDRESS
```

2. Display all available products.

```
SQL> SELECT *
  2  FROM PRODUCT;

PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
     STOCK CATEGORY_ID
---------- ----------
       106 Modern Photo Frame                                    750
        15          1

       107 Premium Acrylic Frame                                 850
        12          2

       108 Large Collage Frame                                  1000
        10          3


PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
     STOCK CATEGORY_ID
---------- ----------
       109 Designer Wall Frame                                   800
        14          4

       110 Classic Table Frame                                   500
        20          5

       101 Classic Wooden Frame                                  600
        20          1


PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
     STOCK CATEGORY_ID
---------- ----------
       102 Elegant Acrylic Frame                                 700
        15          2

       103 Family Collage Frame                                  900
```

```
---------- ----------
       109 Designer Wall Frame                                   800
        14          4

       110 Classic Table Frame                                   500
        20          5

       101 Classic Wooden Frame                                  600
        20          1


PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
     STOCK CATEGORY_ID
---------- ----------
       102 Elegant Acrylic Frame                                 700
        15          2

       103 Family Collage Frame                                  900
        10          3

       104 Decorative Wall Frame                                 650
        12          4


PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
     STOCK CATEGORY_ID
---------- ----------
       105 Table Photo Frame                                     400
        18          5


10 rows selected.
```

3. Retrieve product names and prices only.

```
SQL> SELECT PRODUCT_NAME, PRICE
  2  FROM PRODUCT;

PRODUCT_NAME                                       PRICE
-------------------------------------------------- ----------
Modern Photo Frame                                    750
Premium Acrylic Frame                                 850
Large Collage Frame                                  1000
Designer Wall Frame                                   800
Classic Table Frame                                   500
Classic Wooden Frame                                  600
Elegant Acrylic Frame                                 700
Family Collage Frame                                  900
Decorative Wall Frame                                 650
Table Photo Frame                                     400

10 rows selected.
```

4. Display all orders placed by customers.

```
SQL> SELECT C.CUSTOMER_NAME,
  2         O.ORDER_ID,
  3         O.TOTAL_AMOUNT,
  4         O.ORDER_STATUS
  5  FROM CUSTOMER C
  6  JOIN ORDER_ENTITY O
  7  ON C.CUSTOMER_ID = O.CUSTOMER_ID;

CUSTOMER_N   ORDER_ID TOTAL_AMOUNT
---------- ---------- ------------
ORDER_STATUS
--------------------------------------------------
ARUN             1001         1500
CONFIRMED

PRIYA            1002         2200
SHIPPED

KARTHIK          1003         1800
SHIPPED


CUSTOMER_N   ORDER_ID TOTAL_AMOUNT
---------- ---------- ------------
ORDER_STATUS
--------------------------------------------------
DIVYA            1004         2500
DELIVERED

SURESH           1005         1200
CONFIRMED

MEENA            1006         1700
CONFIRMED


CUSTOMER_N   ORDER_ID TOTAL_AMOUNT
---------- ---------- ------------
ORDER_STATUS
--------------------------------------------------
VIGNESH          1007         1900
PENDING

ANU              1008         2300
SHIPPED

RAHUL            1009         1400
CONFIRMED


CUSTOMER_N   ORDER_ID TOTAL_AMOUNT
---------- ---------- ------------
ORDER_STATUS
--------------------------------------------------
NISHA            1010         2800
DELIVERED


10 rows selected.
```

5. Retrieve payment details:

```
SQL> SELECT *
  2  FROM PAYMENT_ENTITY;

PAYMENT_ID   ORDER_ID PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
---------- ---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
PAY001           1001 20-SEP-26 UPI                  SUCCESS
              1500

PAY002           1002 21-SEP-26 CARD                 SUCCESS
              2200

PAY003           1003 22-SEP-26 NET BANKING          SUCCESS
              1800


PAYMENT_ID   ORDER_ID PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
---------- ---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
PAY004           1004 23-SEP-26 UPI                  SUCCESS
              2500

PAY005           1005 24-SEP-26 CARD                 PENDING
              1200

PAY006           1006 25-SEP-26 UPI                  SUCCESS
              1700


PAYMENT_ID   ORDER_ID PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
---------- ---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
PAY007           1007 26-SEP-26 CARD                 SUCCESS
              1900

PAY008           1008 27-SEP-26 NET BANKING          SUCCESS
              2300

PAY009           1009 28-SEP-26 UPI                  SUCCESS
              1400


PAYMENT_ID   ORDER_ID PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
---------- ---------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
PAY010           1010 29-SEP-26 CARD                 SUCCESS
              2800


10 rows selected.
```

2. Apply Filtering Conditions Using WHERE

1. Find products with price greater than 800

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE PRICE > 800;

PRODUCT_ID PRODUCT_NAME                                       PRICE
---------- -------------------------------------------------- ----------
     STOCK CATEGORY_ID
---------- -----------
       107 Premium Acrylic Frame                               850
        12           2

       108 Large Collage Frame                                1000
        10           3

       103 Family Collage Frame                                900
        10           3
```

2. Display products available in stock.

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE STOCK > 0;

PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
     STOCK CATEGORY_ID
---------- ----------
       106 Modern Photo Frame                                    750
        15           1

       107 Premium Acrylic Frame                                 850
        12           2

       108 Large Collage Frame                                  1000
        10           3


PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
     STOCK CATEGORY_ID
---------- ----------
       109 Designer Wall Frame                                   800
        14           4

       110 Classic Table Frame                                   500
        20           5

       101 Classic Wooden Frame                                  600
        20           1


PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
     STOCK CATEGORY_ID
---------- ----------
       102 Elegant Acrylic Frame                                 700
        15           2

       103 Family Collage Frame                                  900
        10           3

       104 Decorative Wall Frame                                 650
        12           4


PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
     STOCK CATEGORY_ID
---------- ----------
       105 Table Photo Frame                                     400
        18           5


10 rows selected.
```

3. Display completed orders.

```
SQL> SELECT *
  2  FROM ORDER_ENTITY
  3  WHERE ORDER_STATUS = 'DELIVERED';

  ORDER_ID CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT
---------- ----------- --------- ------------
ORDER_STATUS
--------------------------------------------------
      1004         204 23-SEP-26         2500
DELIVERED

      1010         210 29-SEP-26         2800
DELIVERED
```

4. Find products with rating above 4.

```
SQL> SELECT *
  2  FROM REVIEW
  3  WHERE RATING > 4;

REVIEWI CUSTOMERID  PRODUCTID     RATING
------- ---------- ---------- ----------
REVIEWTEST                                         REVIEWDAT
-------------------------------------------------- ---------
REV0001        201        101          5
Excellent product                                  20-SEP-26

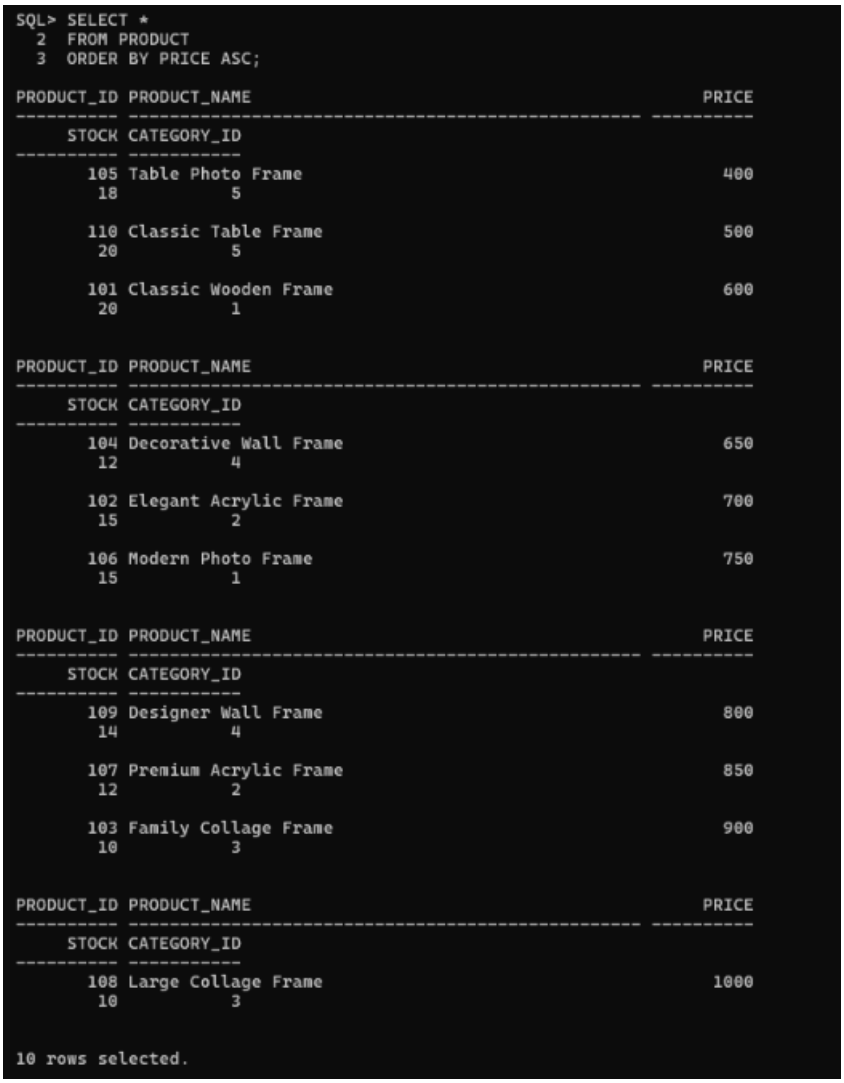
REV0003        203        103          5
Very nice frame                                    22-SEP-26

REV0006        206        106          5
Excellent quality                                  25-SEP-26


REVIEWI CUSTOMERID  PRODUCTID     RATING
------- ---------- ---------- ----------
REVIEWTEST                                         REVIEWDAT
-------------------------------------------------- ---------
REV0009        209        109          5
Beautiful frame                                    28-SEP-26
```

3. Sort Data Using ORDER BY

Tasks: 1. Display products from lowest to highest price.



```
SQL> SELECT *
  2  FROM PRODUCT
  3  ORDER BY PRICE ASC;

PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
     STOCK CATEGORY_ID
---------- ----------
       105 Table Photo Frame                                     400
        18          5

       110 Classic Table Frame                                   500
        20          5

       101 Classic Wooden Frame                                  600
        20          1


PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
     STOCK CATEGORY_ID
---------- ----------
       104 Decorative Wall Frame                                 650
        12          4

       102 Elegant Acrylic Frame                                 700
        15          2

       106 Modern Photo Frame                                    750
        15          1


PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
     STOCK CATEGORY_ID
---------- ----------
       109 Designer Wall Frame                                   800
        14          4

       107 Premium Acrylic Frame                                 850
        12          2

       103 Family Collage Frame                                  900
        10          3


PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
     STOCK CATEGORY_ID
---------- ----------
       108 Large Collage Frame                                  1000
        10          3


10 rows selected.
```

2. Display customers alphabetically.

```
SQL> SELECT *
  2  FROM CUSTOMER
  3  ORDER BY CUSTOMER_NAME ASC;

CUSTOMER_ID CUSTOMER_N EMAIL                PHONE
----------- ---------- -------------------- ----------
ADDRESS
--------------------------------------------------------------------------------
REGISTRAT
---------
        208 ANU        ANU@GMAIL.COM        9876543217
MADURAI
27-SEP-26

        201 ARUN       ARUN@GMAIL.COM       9876543210
CHENNAI
20-SEP-26

CUSTOMER_ID CUSTOMER_N EMAIL                PHONE
----------- ---------- -------------------- ----------
ADDRESS
--------------------------------------------------------------------------------
REGISTRAT
---------

        204 DIVYA      DIVYA@GMAIL.COM      9876543213
THANJAVUR
23-SEP-26

        203 KARTHIK    KARTHIK@GMAIL.COM    9876543212
TRICHY

CUSTOMER_ID CUSTOMER_N EMAIL                PHONE
----------- ---------- -------------------- ----------
ADDRESS
--------------------------------------------------------------------------------
REGISTRAT
---------
22-SEP-26

        206 MEENA      MEENA@GMAIL.COM      9876543215
CHENNAI
25-SEP-26

        210 NISHA      NISHA@GMAIL.COM      9876543219

CUSTOMER_ID CUSTOMER_N EMAIL                PHONE
----------- ---------- -------------------- ----------
ADDRESS
--------------------------------------------------------------------------------
REGISTRAT
---------
TRICHY
29-SEP-26

        202 PRIYA      PRIYA@GMAIL.COM      9876543211
MADURAI
21-SEP-26


CUSTOMER_ID CUSTOMER_N EMAIL                PHONE
----------- ---------- -------------------- ----------
ADDRESS
--------------------------------------------------------------------------------
REGISTRAT
---------
        209 RAHUL      RAHUL@GMAIL.COM      9876543218
```

```
ADDRESS
--------------------------------------------------------------------------------
REGISTRAT
---------

        204 DIVYA      DIVYA@GMAIL.COM      9876543213
THANJAVUR
23-SEP-26

        203 KARTHIK    KARTHIK@GMAIL.COM    9876543212
TRICHY

CUSTOMER_ID CUSTOMER_N EMAIL                PHONE
----------- ---------- -------------------- ----------
ADDRESS
--------------------------------------------------------------------------------
REGISTRAT
---------
22-SEP-26

        206 MEENA      MEENA@GMAIL.COM      9876543215
CHENNAI
25-SEP-26

        210 NISHA      NISHA@GMAIL.COM      9876543219

CUSTOMER_ID CUSTOMER_N EMAIL                PHONE
----------- ---------- -------------------- ----------
ADDRESS
--------------------------------------------------------------------------------
REGISTRAT
---------
TRICHY
29-SEP-26

        202 PRIYA      PRIYA@GMAIL.COM      9876543211
MADURAI
21-SEP-26


CUSTOMER_ID CUSTOMER_N EMAIL                PHONE
----------- ---------- -------------------- ----------
ADDRESS
--------------------------------------------------------------------------------
REGISTRAT
---------
        209 RAHUL      RAHUL@GMAIL.COM      9876543218
CHENNAI
28-SEP-26

        205 SURESH     SURESH@GMAIL.COM     9876543214
SALEM
24-SEP-26

CUSTOMER_ID CUSTOMER_N EMAIL                PHONE
----------- ---------- -------------------- ----------
ADDRESS
--------------------------------------------------------------------------------
REGISTRAT
---------

        207 VIGNESH    VIGNESH@GMAIL.COM    9876543216
COIMBATORE
26-SEP-26

10 rows selected.
```

3. Find top expensive products.

```
SQL> SELECT *
  2  FROM PRODUCT
  3  ORDER BY PRICE DESC;

PRODUCT_ID PRODUCT_NAME                                       PRICE
---------- -------------------------------------------- ----------
     STOCK CATEGORY_ID
---------- -----------
       108 Large Collage Frame                                1000
        10           3

       103 Family Collage Frame                               900
        10           3

       107 Premium Acrylic Frame                              850
        12           2


PRODUCT_ID PRODUCT_NAME                                       PRICE
---------- -------------------------------------------- ----------
     STOCK CATEGORY_ID
---------- -----------
       109 Designer Wall Frame                                800
        14           4

       106 Modern Photo Frame                                 750
        15           1

       102 Elegant Acrylic Frame                              700
        15           2


PRODUCT_ID PRODUCT_NAME                                       PRICE
---------- -------------------------------------------- ----------
     STOCK CATEGORY_ID
---------- -----------
       104 Decorative Wall Frame                              650
        12           4

       101 Classic Wooden Frame                               600
        20           1

       110 Classic Table Frame                                500
        20           5


PRODUCT_ID PRODUCT_NAME                                       PRICE
---------- -------------------------------------------- ----------
     STOCK CATEGORY_ID
---------- -----------
       105 Table Photo Frame                                  400
        18           5


10 rows selected.
```

4. Display latest orders first.

```
SQL> SELECT *
  2  FROM ORDER_ENTITY
  3  ORDER BY ORDER_DATE DESC;

  ORDER_ID CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT
---------- ----------- --------- ------------
ORDER_STATUS
--------------------------------------------------
      1010         210 29-SEP-26         2800
DELIVERED

      1009         209 28-SEP-26         1400
CONFIRMED

      1008         208 27-SEP-26         2300
SHIPPED

  ORDER_ID CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT
---------- ----------- --------- ------------
ORDER_STATUS
--------------------------------------------------
      1007         207 26-SEP-26         1900
PENDING

      1006         206 25-SEP-26         1700
CONFIRMED

      1005         205 24-SEP-26         1200
CONFIRMED

  ORDER_ID CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT
---------- ----------- --------- ------------
ORDER_STATUS
--------------------------------------------------
      1004         204 23-SEP-26         2500
DELIVERED

      1003         203 22-SEP-26         1800
SHIPPED

      1002         202 21-SEP-26         2200
SHIPPED

  ORDER_ID CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT
---------- ----------- --------- ------------
ORDER_STATUS
--------------------------------------------------
      1001         201 20-SEP-26         1500
CONFIRMED

10 rows selected.
```

4. Retrieve Unique Values Using DISTINCT

Tasks: 1. Display unique product categories.

```
SQL> SELECT DISTINCT CATEGORY_NAME
  2  FROM CATEGORY;

CATEGORY_NAME
--------------------------------------------------
Acrylic Photo Frames
Collage Photo Frames
Tabletop Photo Frames
Wall Mounted Frames
Wooden Photo Frames
```

2. Find different payment methods used by customers.

```
SQL> SELECT DISTINCT PAYMENT_MODE
  2  FROM PAYMENT_ENTITY;

PAYMENT_MODE
--------------------
NET BANKING
CARD
UPI
```

5. Search Products Based on Conditions

Tasks: Write SQL queries to:

1. Find products within a specific price range. (Product Between 20,000 to 40,000)

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE PRICE BETWEEN 500 AND 1000;

PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
     STOCK CATEGORY_ID
---------- ----------
       106 Modern Photo Frame                                    750
        15          1

       107 Premium Acrylic Frame                                 850
        12          2

       108 Large Collage Frame                                  1000
        10          3


PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
     STOCK CATEGORY_ID
---------- ----------
       109 Designer Wall Frame                                   800
        14          4

       110 Classic Table Frame                                   500
        20          5

       101 Classic Wooden Frame                                  600
        20          1


PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
     STOCK CATEGORY_ID
---------- ----------
       102 Elegant Acrylic Frame                                 700
        15          2

       103 Family Collage Frame                                  900
        10          3

       104 Decorative Wall Frame                                 650
        12          4


9 rows selected.
```

2. Display products belonging to a particular category.

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE CATEGORY_ID = 1;

PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
     STOCK CATEGORY_ID
---------- ----------
       106 Modern Photo Frame                                    750
        15          1

       101 Classic Wooden Frame                                  600
        20          1
```

Find products currently available in inventory.

```
SQL> SELECT P.PRODUCT_NAME, I.STOCK_QUANTITY, I.STOCK_STATUS
  2  FROM PRODUCT P
  3  JOIN INVENTORY I
  4  ON P.PRODUCT_ID = I.PRODUCT_ID
  5  WHERE I.STOCK_STATUS = 'AVAILABLE';

PRODUCT_NAME                                       STOCK_QUANTITY
-------------------------------------------------- --------------
STOCK_STATUS
---------------
Classic Wooden Frame                                            5
AVAILABLE

Elegant Acrylic Frame                                           5
AVAILABLE

Family Collage Frame                                            6
AVAILABLE


PRODUCT_NAME                                       STOCK_QUANTITY
-------------------------------------------------- --------------
STOCK_STATUS
---------------
Decorative Wall Frame                                           9
AVAILABLE
```

3. Search products using product name.

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE PRODUCT_NAME LIKE '%Frame%';

PRODUCT_ID PRODUCT_NAME                                      PRICE
---------- -------------------------------------------------- ----------
     STOCK CATEGORY_ID
---------- -----------
       106 Modern Photo Frame                                  750
        15           1

       107 Premium Acrylic Frame                               850
        12           2

       108 Large Collage Frame                                1000
        10           3


PRODUCT_ID PRODUCT_NAME                                      PRICE
---------- -------------------------------------------------- ----------
     STOCK CATEGORY_ID
---------- -----------
       109 Designer Wall Frame                                 800
        14           4

       110 Classic Table Frame                                 500
        20           5

       101 Classic Wooden Frame                                600
        20           1


PRODUCT_ID PRODUCT_NAME                                      PRICE
---------- -------------------------------------------------- ----------
     STOCK CATEGORY_ID
---------- -----------
       102 Elegant Acrylic Frame                               700
        15           2

       103 Family Collage Frame                                900
        10           3

       104 Decorative Wall Frame                               650
        12           4


PRODUCT_ID PRODUCT_NAME                                      PRICE
---------- -------------------------------------------------- ----------
     STOCK CATEGORY_ID
---------- -----------
       105 Table Photo Frame                                   400
        18           5


10 rows selected.
```

4. Find low-stock products.

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE STOCK < 15;

PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
     STOCK CATEGORY_ID
---------- ----------
       107 Premium Acrylic Frame                                 850
        12          2

       108 Large Collage Frame                                  1000
        10          3

       109 Designer Wall Frame                                   800
        14          4


PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
     STOCK CATEGORY_ID
---------- ----------
       103 Family Collage Frame                                  900
        10          3

       104 Decorative Wall Frame                                 650
        12          4
```

5. Retrieve Customer and Product Information

   Tasks: 1. Display customer details with their orders.

```
SQL> SELECT C.*, O.ORDER_ID, O.ORDER_DATE,
  2         O.TOTAL_AMOUNT, O.ORDER_STATUS
  3  FROM CUSTOMER C
  4  JOIN ORDER_ENTITY O
  5  ON C.CUSTOMER_ID = O.CUSTOMER_ID;

CUSTOMER_ID CUSTOMER_N EMAIL                    PHONE
----------- ---------- -------------------- ----------
ADDRESS
--------------------------------------------------------------------------------
REGISTRAT   ORDER_ID ORDER_DAT TOTAL_AMOUNT
--------- ---------- --------- ------------
ORDER_STATUS
--------------------------------------------------
        201 ARUN       ARUN@GMAIL.COM       9876543210
CHENNAI
20-SEP-26       1001 20-SEP-26         1500
CONFIRMED


CUSTOMER_ID CUSTOMER_N EMAIL                    PHONE
----------- ---------- -------------------- ----------
ADDRESS
--------------------------------------------------------------------------------
REGISTRAT   ORDER_ID ORDER_DAT TOTAL_AMOUNT
--------- ---------- --------- ------------
ORDER_STATUS
--------------------------------------------------
        202 PRIYA      PRIYA@GMAIL.COM      9876543211
MADURAI
21-SEP-26       1002 21-SEP-26         2200
SHIPPED


CUSTOMER_ID CUSTOMER_N EMAIL                    PHONE
----------- ---------- -------------------- ----------
ADDRESS
--------------------------------------------------------------------------------
REGISTRAT   ORDER_ID ORDER_DAT TOTAL_AMOUNT
--------- ---------- --------- ------------
ORDER_STATUS
--------------------------------------------------
        203 KARTHIK    KARTHIK@GMAIL.COM    9876543212
TRICHY
22-SEP-26       1003 22-SEP-26         1800
SHIPPED


CUSTOMER_ID CUSTOMER_N EMAIL                    PHONE
----------- ---------- -------------------- ----------
ADDRESS
--------------------------------------------------------------------------------
REGISTRAT   ORDER_ID ORDER_DAT TOTAL_AMOUNT
--------- ---------- --------- ------------
ORDER_STATUS
--------------------------------------------------
        204 DIVYA      DIVYA@GMAIL.COM      9876543213
THANJAVUR
23-SEP-26       1004 23-SEP-26         2500
DELIVERED
```

```
CUSTOMER_ID CUSTOMER_N EMAIL                    PHONE
----------- ---------- -------------------- ----------
ADDRESS
--------------------------------------------------------------------------------
REGISTRAT   ORDER_ID ORDER_DAT TOTAL_AMOUNT
--------- ---------- --------- ------------
ORDER_STATUS
--------------------------------------------------
        207 VIGNESH    VIGNESH@GMAIL.COM    9876543216
COIMBATORE
26-SEP-26       1007 26-SEP-26         1900
PENDING


CUSTOMER_ID CUSTOMER_N EMAIL                    PHONE
----------- ---------- -------------------- ----------
ADDRESS
--------------------------------------------------------------------------------
REGISTRAT   ORDER_ID ORDER_DAT TOTAL_AMOUNT
--------- ---------- --------- ------------
ORDER_STATUS
--------------------------------------------------
        208 ANU        ANU@GMAIL.COM        9876543217
MADURAI
27-SEP-26       1008 27-SEP-26         2300
SHIPPED


CUSTOMER_ID CUSTOMER_N EMAIL                    PHONE
----------- ---------- -------------------- ----------
ADDRESS
--------------------------------------------------------------------------------
REGISTRAT   ORDER_ID ORDER_DAT TOTAL_AMOUNT
--------- ---------- --------- ------------
ORDER_STATUS
--------------------------------------------------
        209 RAHUL      RAHUL@GMAIL.COM      9876543218
CHENNAI
28-SEP-26       1009 28-SEP-26         1400
CONFIRMED


CUSTOMER_ID CUSTOMER_N EMAIL                    PHONE
----------- ---------- -------------------- ----------
ADDRESS
--------------------------------------------------------------------------------
REGISTRAT   ORDER_ID ORDER_DAT TOTAL_AMOUNT
--------- ---------- --------- ------------
ORDER_STATUS
--------------------------------------------------
        210 NISHA      NISHA@GMAIL.COM      9876543219
TRICHY
29-SEP-26       1010 29-SEP-26         2800
DELIVERED


10 rows selected.
```

2. Display product details with category information.

```
SQL> SELECT P.*, C.CATEGORY_NAME
  2  FROM PRODUCT P
  3  JOIN CATEGORY C
  4  ON P.CATEGORY_ID = C.CATEGORY_ID;

PRODUCT_ID PRODUCT_NAME                                           PRICE
---------- -------------------------------------------------- ----------
     STOCK CATEGORY_ID CATEGORY_NAME
---------- ----------- --------------------------------------------------
       106 Modern Photo Frame                                        750
        15           1 Wooden Photo Frames

       107 Premium Acrylic Frame                                     850
        12           2 Acrylic Photo Frames

       108 Large Collage Frame                                      1000
        10           3 Collage Photo Frames


PRODUCT_ID PRODUCT_NAME                                           PRICE
---------- -------------------------------------------------- ----------
     STOCK CATEGORY_ID CATEGORY_NAME
---------- ----------- --------------------------------------------------
       109 Designer Wall Frame                                       800
        14           4 Wall Mounted Frames

       110 Classic Table Frame                                       500
        20           5 Tabletop Photo Frames

       101 Classic Wooden Frame                                      600
        20           1 Wooden Photo Frames


PRODUCT_ID PRODUCT_NAME                                           PRICE
---------- -------------------------------------------------- ----------
     STOCK CATEGORY_ID CATEGORY_NAME
---------- ----------- --------------------------------------------------
       102 Elegant Acrylic Frame                                     700
        15           2 Acrylic Photo Frames

       103 Family Collage Frame                                      900
        10           3 Collage Photo Frames

       104 Decorative Wall Frame                                     650
        12           4 Wall Mounted Frames


PRODUCT_ID PRODUCT_NAME                                           PRICE
---------- -------------------------------------------------- ----------
     STOCK CATEGORY_ID CATEGORY_NAME
---------- ----------- --------------------------------------------------
       105 Table Photo Frame                                         400
        18           5 Tabletop Photo Frames


10 rows selected.
```

3. Find customers who purchased a specific product.

```
SQL> SELECT DISTINCT
  2         C.CUSTOMER_ID,
  3         C.CUSTOMER_NAME,
  4         C.EMAIL,
  5         C.PHONE
  6  FROM CUSTOMER C
  7  JOIN ORDER_ENTITY O
  8      ON C.CUSTOMER_ID = O.CUSTOMER_ID
  9  JOIN ORDER_DETAILS_ENTITY OD
 10      ON O.ORDER_ID = OD.ORDER_ID
 11  JOIN PRODUCT P
 12      ON OD.PRODUCT_ID = P.PRODUCT_ID
 13  WHERE P.PRODUCT_ID = 101;

CUSTOMER_ID CUSTOMER_N EMAIL                    PHONE
----------- ---------- -------------------- ----------
        206 MEENA      MEENA@GMAIL.COM      9876543215
        201 ARUN       ARUN@GMAIL.COM       9876543210
```

4. Display products purchased by each customer.

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         P.PRODUCT_NAME
  4  FROM CUSTOMER C
  5  JOIN ORDER_ENTITY O
  6      ON C.CUSTOMER_ID = O.CUSTOMER_ID
  7  JOIN ORDER_DETAILS_ENTITY OD
  8      ON O.ORDER_ID = OD.ORDER_ID
  9  JOIN PRODUCT P
 10      ON OD.PRODUCT_ID = P.PRODUCT_ID;

CUSTOMER_ID CUSTOMER_N PRODUCT_NAME
----------- ---------- --------------------------------------------------
        206 MEENA      Classic Wooden Frame
        201 ARUN       Classic Wooden Frame
        207 VIGNESH    Elegant Acrylic Frame
        202 PRIYA      Elegant Acrylic Frame
        208 ANU        Family Collage Frame
        203 KARTHIK    Family Collage Frame
        209 RAHUL      Decorative Wall Frame
        204 DIVYA      Decorative Wall Frame
        210 NISHA      Table Photo Frame
        205 SURESH     Table Photo Frame

10 rows selected.
```

6. Apply Multiple Filtering Conditions

Tasks: 1. Find electronics products costing more than ₹500

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE CATEGORY_ID = 1
  4  AND PRICE > 500;

PRODUCT_ID PRODUCT_NAME                                       PRICE
---------- -------------------------------------------------- ----------
     STOCK CATEGORY_ID
---------- -----------
       106 Modern Photo Frame                                    750
        15           1

       101 Classic Wooden Frame                                  600
        20           1
```

2. Search products containing the word "frame".

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE PRODUCT_NAME LIKE '%Frame%';

PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
     STOCK CATEGORY_ID
---------- ----------
       106 Modern Photo Frame                                    750
        15           1

       107 Premium Acrylic Frame                                 850
        12           2

       108 Large Collage Frame                                  1000
        10           3


PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
     STOCK CATEGORY_ID
---------- ----------
       109 Designer Wall Frame                                   800
        14           4

       110 Classic Table Frame                                   500
        20           5

       101 Classic Wooden Frame                                  600
        20           1


PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
     STOCK CATEGORY_ID
---------- ----------
       102 Elegant Acrylic Frame                                 700
        15           2

       103 Family Collage Frame                                  900
        10           3

       104 Decorative Wall Frame                                 650
        12           4


PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
     STOCK CATEGORY_ID
---------- ----------
       105 Table Photo Frame                                     400
        18           5


10 rows selected.
```

Find Orders Placed Within a Specific Date Range

```
SQL> SELECT *
  2  FROM ORDER_ENTITY
  3  WHERE ORDER_DATE BETWEEN
  4  TO_DATE('20-09-2026', 'DD-MM-YYYY')
  5  AND TO_DATE('25-09-2026', 'DD-MM-YYYY');

  ORDER_ID CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT
---------- ----------- --------- ------------
ORDER_STATUS
--------------------------------------------------
      1001         201 20-SEP-26         1500
CONFIRMED

      1002         202 21-SEP-26         2200
SHIPPED

      1003         203 22-SEP-26         1800
SHIPPED


  ORDER_ID CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT
---------- ----------- --------- ------------
ORDER_STATUS
--------------------------------------------------
      1004         204 23-SEP-26         2500
DELIVERED

      1005         205 24-SEP-26         1200
CONFIRMED

      1006         206 25-SEP-26         1700
CONFIRMED


6 rows selected.
```

7. Generate Basic Business Reports

   Report 1: Product Availability Report Display:

   Product name

   Price

   Stock quantity

Availability status

```
SQL> SELECT PRODUCT_NAME,
  2         PRICE,
  3         STOCK AS STOCK_QUANTITY,
  4         CASE
  5             WHEN STOCK > 0 THEN 'AVAILABLE'
  6             ELSE 'OUT OF STOCK'
  7         END AS AVAILABILITY_STATUS
  8  FROM PRODUCT;

PRODUCT_NAME                                       PRICE STOCK_QUANTITY
-------------------------------------------------- ----------  --------------
AVAILABILITY
------------
Modern Photo Frame                                   750              15
AVAILABLE

Premium Acrylic Frame                                850              12
AVAILABLE

Large Collage Frame                                 1000              10
AVAILABLE


PRODUCT_NAME                                       PRICE STOCK_QUANTITY
-------------------------------------------------- ----------  --------------
AVAILABILITY
------------
Designer Wall Frame                                  800              14
AVAILABLE

Classic Table Frame                                  500              20
AVAILABLE

Classic Wooden Frame                                 600              20
AVAILABLE


PRODUCT_NAME                                       PRICE STOCK_QUANTITY
-------------------------------------------------- ----------  --------------
AVAILABILITY
------------
Elegant Acrylic Frame                                700              15
AVAILABLE

Family Collage Frame                                 900              10
AVAILABLE

Decorative Wall Frame                                650              12
AVAILABLE


PRODUCT_NAME                                       PRICE STOCK_QUANTITY
-------------------------------------------------- ----------  --------------
AVAILABILITY
------------
Table Photo Frame                                    400              18
AVAILABLE


10 rows selected.
```

Report 2: Customer Report Generate:

Total number of customers

```
SQL> SELECT COUNT(*) AS TOTAL_CUSTOMERS
  2  FROM CUSTOMER;

TOTAL_CUSTOMERS
---------------
             10
```

New customer registrations

```
SQL> SELECT *
  2  FROM CUSTOMER
  3  WHERE REGISTRATION_DATE >=
  4  TO_DATE('25-09-2026', 'DD-MM-YYYY');

CUSTOMER_ID CUSTOMER_N EMAIL                    PHONE
----------- ---------- -------------------- ----------
ADDRESS
--------------------------------------------------------------------------------
REGISTRAT
---------
        206 MEENA      MEENA@GMAIL.COM      9876543215
CHENNAI
25-SEP-26

        207 VIGNESH    VIGNESH@GMAIL.COM    9876543216
COIMBATORE
26-SEP-26

CUSTOMER_ID CUSTOMER_N EMAIL                    PHONE
----------- ---------- -------------------- ----------
ADDRESS
--------------------------------------------------------------------------------
REGISTRAT
---------

        208 ANU        ANU@GMAIL.COM        9876543217
MADURAI
27-SEP-26

        209 RAHUL      RAHUL@GMAIL.COM      9876543218
CHENNAI

CUSTOMER_ID CUSTOMER_N EMAIL                    PHONE
----------- ---------- -------------------- ----------
ADDRESS
--------------------------------------------------------------------------------
REGISTRAT
---------
28-SEP-26

        210 NISHA      NISHA@GMAIL.COM      9876543219
TRICHY
29-SEP-26
```

Report 3: Order Report Display: Total orders

```
SQL> SELECT COUNT(*) AS TOTAL_ORDERS
  2  FROM ORDER_ENTITY;

TOTAL_ORDERS
------------
          10
```

Completed orders

```
SQL> SELECT COUNT(*) AS COMPLETED_ORDERS
  2  FROM ORDER_ENTITY
  3  WHERE ORDER_STATUS = 'DELIVERED';

COMPLETED_ORDERS
----------------
               2
```

Pending orders

```
SQL> SELECT COUNT(*) AS PENDING_ORDERS
  2  FROM ORDER_ENTITY
  3  WHERE ORDER_STATUS = 'PENDING';

PENDING_ORDERS
--------------
             1
```

Cancelled orders

```
SQL> SELECT COUNT(*) AS CANCELLED_ORDERS
  2  FROM ORDER_ENTITY
  3  WHERE ORDER_STATUS = 'CANCELLED';

CANCELLED_ORDERS
----------------
               0
```

Report 4: Product Performance Report

Find:

Highest-priced products

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE PRICE = (
  4      SELECT MAX(PRICE)
  5      FROM PRODUCT
  6  );

PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
     STOCK CATEGORY_ID
---------- ----------
       108 Large Collage Frame                                  1000
        10           3
```

Most reviewed products

```
SQL> SELECT P.PRODUCT_NAME,
  2         COUNT(R.REVIEWID) AS REVIEW_COUNT
  3  FROM PRODUCT P
  4  JOIN REVIEW R
  5  ON P.PRODUCT_ID = R.PRODUCTID
  6  GROUP BY P.PRODUCT_NAME
  7  ORDER BY REVIEW_COUNT DESC;

PRODUCT_NAME                                       REVIEW_COUNT
-------------------------------------------------- ------------
Decorative Wall Frame                                         1
Premium Acrylic Frame                                         1
Large Collage Frame                                           1
Modern Photo Frame                                            1
Classic Table Frame                                           1
Classic Wooden Frame                                          1
Elegant Acrylic Frame                                         1
Designer Wall Frame                                           1
Family Collage Frame                                          1
Table Photo Frame                                             1

10 rows selected.
```

Available products

```
SQL> SELECT *
  2  FROM PRODUCT
  3  WHERE STOCK > 0;

PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
     STOCK CATEGORY_ID
---------- ----------
       106 Modern Photo Frame                                   750
        15           1

       107 Premium Acrylic Frame                                850
        12           2

       108 Large Collage Frame                                 1000
        10           3


PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
     STOCK CATEGORY_ID
---------- ----------
       109 Designer Wall Frame                                  800
        14           4

       110 Classic Table Frame                                  500
        20           5

       101 Classic Wooden Frame                                 600
        20           1


PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
     STOCK CATEGORY_ID
---------- ----------
       102 Elegant Acrylic Frame                                700
        15           2

       103 Family Collage Frame                                 900
        10           3

       104 Decorative Wall Frame                                650
        12           4


PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
     STOCK CATEGORY_ID
---------- ----------
       105 Table Photo Frame                                    400
        18           5


10 rows selected.
```